Lösung: Logarithmus- und Potenzgesetze

a)

$$\frac{2}{e^{2\cdot \underbrace{\ln(\sqrt{3})}_{1/2\cdot\ln(3)}}} + e^{\overbrace{1/2\cdot\ln(4)}^{=\ln(4^{1/2})}} \cdot e^{\overbrace{-\ln(3)-\ln(2)}^{=-\ln(3\cdot 2)}}$$

$$= \frac{2}{e^{\ln(3)}} + \frac{e^{\ln(2)}}{e^{\ln(6)}} = \frac{2}{3} + \frac{2}{6} = \frac{3}{3} = \underline{\underline{1}}$$

b)

$$e^{-\ln(\sqrt{2})} - \ln\left(\sqrt{e^{\sqrt{2}}}\right)$$

$$= \frac{1}{e^{\ln(\sqrt{2})}} - \frac{1}{2}\ln\left(e^{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} - \frac{1}{2}\sqrt{2}$$

$$= \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = \underline{\underline{0}}$$